

EEG Motor Imagery Signals Classification using Maximum Overlap Wavelet Transform and Support Vector Machine

Cesar E. Hernández-González*, Juan M Ramírez-Cortés*, Pilar Gómez-Gil,[†]
 José Rangel-Magdaleno* Hayde Peregrina-Barreto[†] and Israel Cruz-Vega*

*Electronic Department

[†]Computer Science Department

Instituto Nacional de Astrofísica Óptica y Electrónica

E-mail{cesar.e.hdz@inaoep.mx, jmram@inaoep.mx, pgomez@inaoep.mx,
 jrangel@inaoep.mx, hayde@inaoep.mx, icruzv@inaoep.mx}

Abstract—A BCI system (Brain-Computer Interface) aims to the interpretation of brain signals perceived through electroencephalography (EEG) sensors in order to allow the user interaction with the environment through specific actions. In this paper we present an experiment of EEG signal classification under the motor imagery paradigm using two feature extraction methods for comparison purposes: discrete wavelet transform (DWT) and maximum overlap discrete wavelet transform (MODWT). The feature vectors are fed into a support vector machine (SVM) classification system. The results obtained show an accuracy of 98.81% in average.

Key words: Electroencephalography (EEG); brain computer interfaces (BCI); discrete wavelet transform (DWT); maximum overlap discrete wavelet transform (MODWT); support vector machine (SVM).

I. INTRODUCTION

In the last decades, the complex processes involved in the brain activity have been thoroughly studied aiming to provide an alternate communication path through signals perceived with EEG electrodes for brain-computer interface (BCI) applications [1]. BCI applications range from devices control using sensorimotor signals [2], control cursor movement [3], or BCI-based speller systems [4], just to mention some. EEG signals show the post-synaptic potentials produced inside the brain, and reflected to the scalp. Electroencephalography provides a noninvasive measure of cortical activity adequate for probing human brain dynamics with a good balance between spatial and temporal resolution. BCI is a term broadly used to describe a system which translates the electrical signals generated from cognitive processes into control signals for a variety of applications, such as computers, home appliances, or mechanical prostheses. There are several signals associated with specific EEG-based paradigms which can be used to develop BCI systems, such as slow cortical potential [5], μ and β rhythms [6], motor imagery [7], visually evoked potentials [8], P300 evoked potentials [9], and combinations of them [10-12]. Among these paradigms, motor imagery (MI)

defined as a mental enforcement involved in the movement of some part of the body without any muscular activity has received considerable attention due to its pertinence to relate a control action with movement intent. [7,10,13,14]. In the context of non-stationary signal analysis, such as the EEG signals involved in BCI applications, wavelet analysis provides an excellent tool for extracting significant information about the time-frequency behavior of the signal, with a good tradeoff on performance and computational complexity. In this work, an experiment on motor imagery signals using wavelet decomposition, and support vector machine classification system, is presented. A novel technique using maximum overlap discrete wavelet transform over pre-processed signals, statistical feature extraction of the extracted bands, and support vector machine classification provides high accuracy results. A series of experiments was conducted using several kernels such as linear, quadratic and cubic in the SVM classifier. The EEG data used in this work was obtained from the BCI Competition IV database, available to the international community for academic and research purposes [15,16]. The rest of the paper is organized as follows: Section 2 summarizes the experiment protocol involved in the database construction, Section 3 presents the methodology proposed in this work: wavelet decomposition, feature extraction, feature reduction and classification. Section 4 presents results of the classification in the form of confusion matrix, and in Section 5 some concluding remarks are discussed.

II. DATABASE DESCRIPTION

The EEG data used in this work was obtained from the public repository of the BCI Competition IV available to the international community for academic and research purposes. It corresponds to experiments with imagined left and right hand, a foot and tongue movements. A detailed description about the signal acquisition and experimental paradigm can be found in [15,16]. Briefly, the database consists of four different tasks of motor imagery of the left hand (class 1),

right hand (class 2), both feet (class 3) and tongue (class 4). Two recording sessions were carried out in different days. Each session consisted of 6 runs separated by short rests. Each run consisted of 48 trials (12 for each class). Figure 1 shows the timeline of the experiment used for data collection.

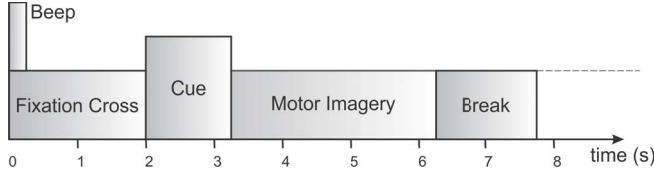


Fig. 1. Timing scheme of the paradigm.

The electroencephalography session was recorded with 22 Ag/AgCl electrodes, located under 10-20 EEG system as shown in figure 2.

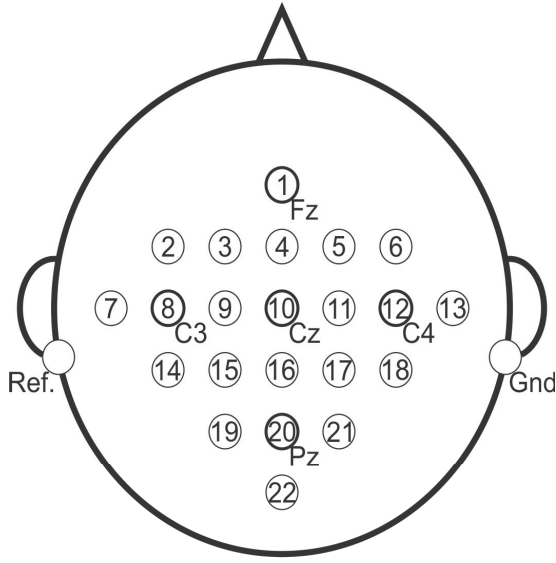


Fig. 2. Ag/AgCL electrodes montage.

The signals were recorded using right mastoid as ground and left mastoid as reference. Signal was sampled with 250 Hz and processed by a band pass filter between .5 and 100 Hz. Amplifier sensitivity was set in 100 μV . A 60 Hz notch filter was used in order to reduce power line noise.

III. METHODOLOGY

Figure 3 shows a block diagram of the EEG signal analysis procedure. The whole process was implemented in Matlab2016b. Each block is described in detail as follows.

A. Wavelet decomposition

Several spectral techniques in the context of BCI systems have been proposed, such as traditional Fourier analysis [17], Empirical Mode Decomposition [18], and Wavelet transform in diverse modalities, with the last one reported having a very good performance when it is combined with higher order statistics of the extracted bands [19,20]. The Discrete Wavelet

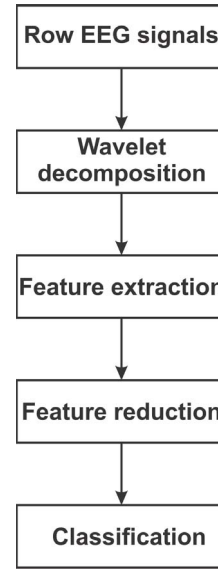


Fig. 3. Block diagram of EEG analysis.

Transform (DWT) is a transformation used to analyze the temporal and spectral properties of non-stationary signals. DWT is defined by the following equation:

$$W(j, k) = \sum_j \sum_k f(x) 2^{-\frac{j}{2}} \psi(2^{-j}x - k) \quad (1)$$

The set of functions $\psi_{j,k}(n)$ is referred to as the family of wavelets derived from $\psi(n)$, which is a time function with finite energy and fast decay called the mother wavelet. The basis of the wavelet space corresponds then, to the orthonormal functions obtained from the mother wavelet after scale and translation operations. The definition indicates the projection of the input signal into the wavelet space through the inner product. Sub-band coding is a popular pyramidal algorithm used to implement DWT in terms of multi-rate filter banks. DWT can be viewed as a filter bank with octave spacing between filters. Each sub-band contains half the samples of the neighboring higher frequency sub-band. In the pyramidal algorithm the signal is analyzed using progressive partition of spectral bands scaling resolution in each decomposition level. This is achieved by successive high-pass and low-pass filtering of the time signal and down-sampling by two. The maximum overlap discrete wavelet transform is a modification of a DWT in which subsampling has been omitted. In consequence, the orthogonality property of the coefficient sets is lost but the sum of the obtained components reconstructs the original signal. MODWT is shift invariant, and it does not generate phase shifts within detail and approximation components.

In the experiment described in this paper, Wavelet analysis was used to divide EEG signals into the spectral bands alpha (α), beta (β), delta (δ) and theta (θ), which contain information related to the brain activity [21, 22]. Delta waves (δ) can be found between 0-4 Hz and are generated principally during deep meditation and deep sleep. Theta waves (θ) can be found in 4-8 Hz and are present in light sleep, meditation,

learning and memorization. Alpha waves (α) are located at 8-12 Hz frequency band and can be generated during fluid thoughts and when the brain rest. Beta waves (β) can be found between 12-32 Hz and are generated when the brain is in alert to cognitive task or outside world. Mu rhythm is located at 8-13 Hz, and it is associated with the planning and execution of voluntary movements of the different parts of the body [21, 22]. EEG raw signals were resampled to 128 Hz in order to adequate them before to wavelet decomposition. Spectral analysis decomposition was carried out using both techniques for comparison purposes: discrete wavelet transform (DWT), and maximum overlap discrete wavelet transform (MODWT). In both cases a Haar mother wavelet was used. Figure 4 shows an EEG signal decomposed in 4 levels by a maximum overlap discrete wavelet transform (MODWT).

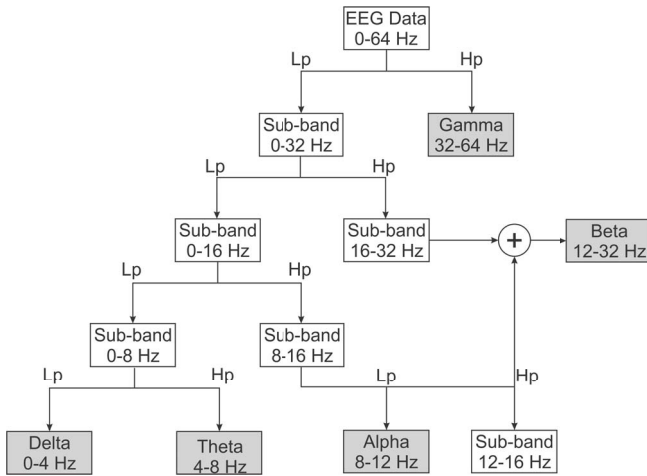


Fig. 4. Decomposition of the EEG signal in 4 levels by MODWT, with each sub-band of interest highlighted

Figure 5 shows an example of each band ($\alpha, \beta, \delta, \theta$) after decomposition

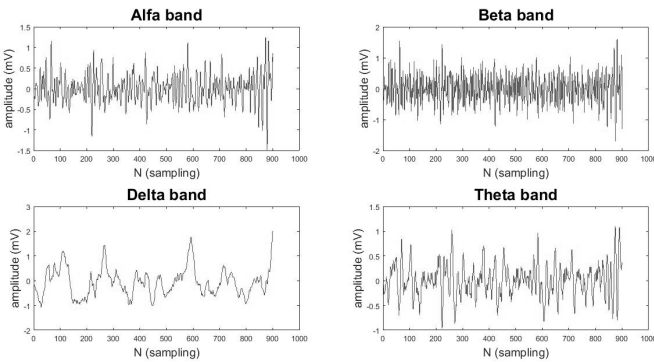


Fig. 5. Graphics of bands ($\alpha, \beta, \delta, \theta$) after MODWT decomposition

B. Feature extraction

The process of feature extraction relies on statistical indicators computed from each sub-band obtained after wavelet decomposition over a sliding window of 25 samples over

the EEG signal. The first two statistical indicators are mean and variance, as expressed in Equations 2-3. The third used indicator is the median, defined as the value of the central position variable in a set of ordered data. The fourth indicator used to generate the feature vectors is Shannon entropy, which represents a measure of uncertainty or disorder in a random signal, and it is expressed in Equation 4.

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \quad (2)$$

$$\sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 \quad (3)$$

$$H(x) = - \sum_{i=1}^n x_i \log_2(x_i) \quad (4)$$

C. Classification

The classification stage was carried out by a support vector machine (SVM) with several kernels for comparison purposes. SVM is a supervised learning algorithm consisting of a statistical learning method based on a structural risk minimization procedure. The basic concept of the algorithm is a mapping of the input space into a higher dimensional feature space in which linear separation is possible [23, 24]. Mapping can be done either linearly or non-linearly, according to the used kernel function. In the new feature space, the SVM constructs separating hyperplanes that are optimal in the sense that the classes are separated with the largest margin and minimum classification error. The optimal hyperplane can be written as a combination of a few feature points, which are called the support vectors of the optimal hyper plane. In this work we use linear, quadratic and cubic functions as mapping kernel, for comparison purposes.

IV. RESULTS

The proposed methodology was tested using two approaches based on discrete wavelet transform and maximum overlap wavelet transform, with four-level decomposition in both cases using a similar feature extraction procedure. A sliding window of 25 points is used to carry out the wavelet-based feature extraction process over the EEG signals, followed by statistical analysis of the information contained in each decomposition band. The process is repeated for each class with a data organization as represented in Figure 6.

The system performance was evaluated using a 5-fold cross validation scheme in both experiments. K-fold cross validation is a technique in which all data is divided into k equal parts, and then, one part is used for training and the rest for evaluation. The process is repeated k times using in each case different database groups as training data, and registering the average of the obtained results. Table I shows the average classification rate obtained with the proposed systems previously described, using SVM with three different kernels: linear, quadratic, and cubic. It can be noticed that the best case in average was obtained with MODWT decomposition and SVM

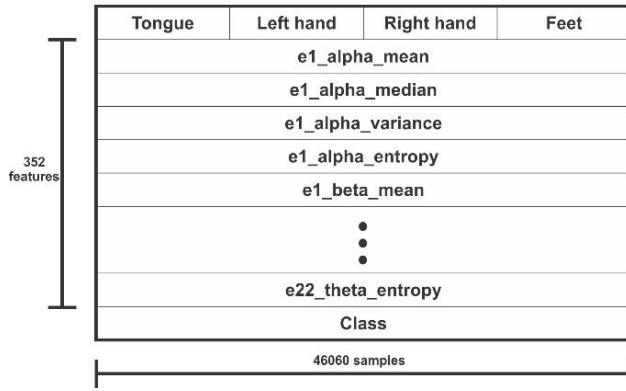


Fig. 6. Data organization.

cubic kernel. Figure 7 shows an example of the confusion matrix of one subject. Results show an excellent classification rate above 80% in all cases. The best performance was obtained through a wavelet decomposition using a MODWT, and a SVM classification with cubic Kernel exhibiting an accuracy of 98.81% in average. A comparison with results obtained from different feature extraction methods reported in the literature shows a very competitive performance.

TABLE I
RESULTS ON CLASSIFICATION RATE.

Decomposition	Kernel	Classification Rate
MODWT	Lineal	81.17%
MODWT	Quadratic	99.72%
MODWT	Cubic	99.80%
DWT	Lineal	84.88%
DWT	Quadratic	99.58%
DWT	Cubic	99.72%

V. CONCLUSIONS

In this paper an EEG signal classification approach under the motor imagery paradigm in the context of BCI systems has been presented. A feature extraction process based on statistical analysis applied to the sub-bands obtained from a Maximum Overlap Discrete Wavelet decomposition provided an excellent class separability. Support vector machine classification provided an overall accuracy of 98.81% in average, using a cubic kernel in the best case. Further experiments applying this methodology to robot control with basic movements are currently in progress.

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977	8	4	11	Tongue
10	953	25	12	Right Hand
6	14	974	6	Left Hand
6	14	9	971	Feet
Tongue	Right Hand	Left Hand	Feet	96.87%

Fig. 7. Example of a confusion matrix obtained with a subject using MODWT and SVM with cubic kernel.

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